

Lab: Steganography

Estimated Time: 45 minutes

Overview

This lab will demonstrate how to perform steganography—hiding information within other files—using tools available on Linux. You will learn to hide and retrieve hidden messages within images and audio files.

Learning objectives

By the end of this lab, you will be able to:

- Embed hidden messages within image or audio files using steganography tools
- Extract concealed data from steganographic files.
- Identify the role of steganography in data concealment and secure communication

Prerequisites (Optional)

- A Linux system with steghide should be installed.
- Basic understanding of steganography.

Initializing the Lab Environment

Follow the below steps to pen the new terminal.

1. Click **Terminal**.

FileEditSelectionViewGoRunTerminalHelp

←→

☐

SKILLS ...

> DATABASES

> BIG DATA

> CLOUD

> EMBEDDABLE AI

> OTHER

Launch Appli...

Welcome X

Cloud IDE

Version 1.7.4

VS Code API Versio

What is Cloud

Cloud IDE is a full

Environment (IDE

debug your code

Getting Started

On your left,you'll find the instructions for the tasks you need to complete in this lab. Follow them step-by-step to achieve the objectives of this lab. You can also click the chat button to ask for help from Tai, your AI Teaching Assistant at any time.

On your right (here),is the Cloud IDE tool, a fully-featured Integrated Development Environment (IDE) that you'll be using to write, run, and debug your code.

Here are some key components of the Cloud IDE tool:

- **File Explorer:** Navigate through your files and directories. [Click HERE to open the File Explorer.](#)
- **Terminal:** Run your code and see the output here. [Click HERE to open a New Terminal.](#)
- **Skills Network Toolbox:** Access various technologies like databases and Embeddable AI with just a click. [Click HERE to toggle the Skills Network Toolbox.](#)

☒ Show welcome page on startup

New TerminalCtrl+Shift+`

New Terminal (With Profile)...

Choose Default Profile...

Run Task...

Run Build Task

Run Test Task

Rerun Last TaskCtrl+Shift+K

Show Running Tasks...

Restart Running Task...

Terminate Task...

Attach Task...

Configure Tasks...



SkillsNetwork

Open Workspace

Recent

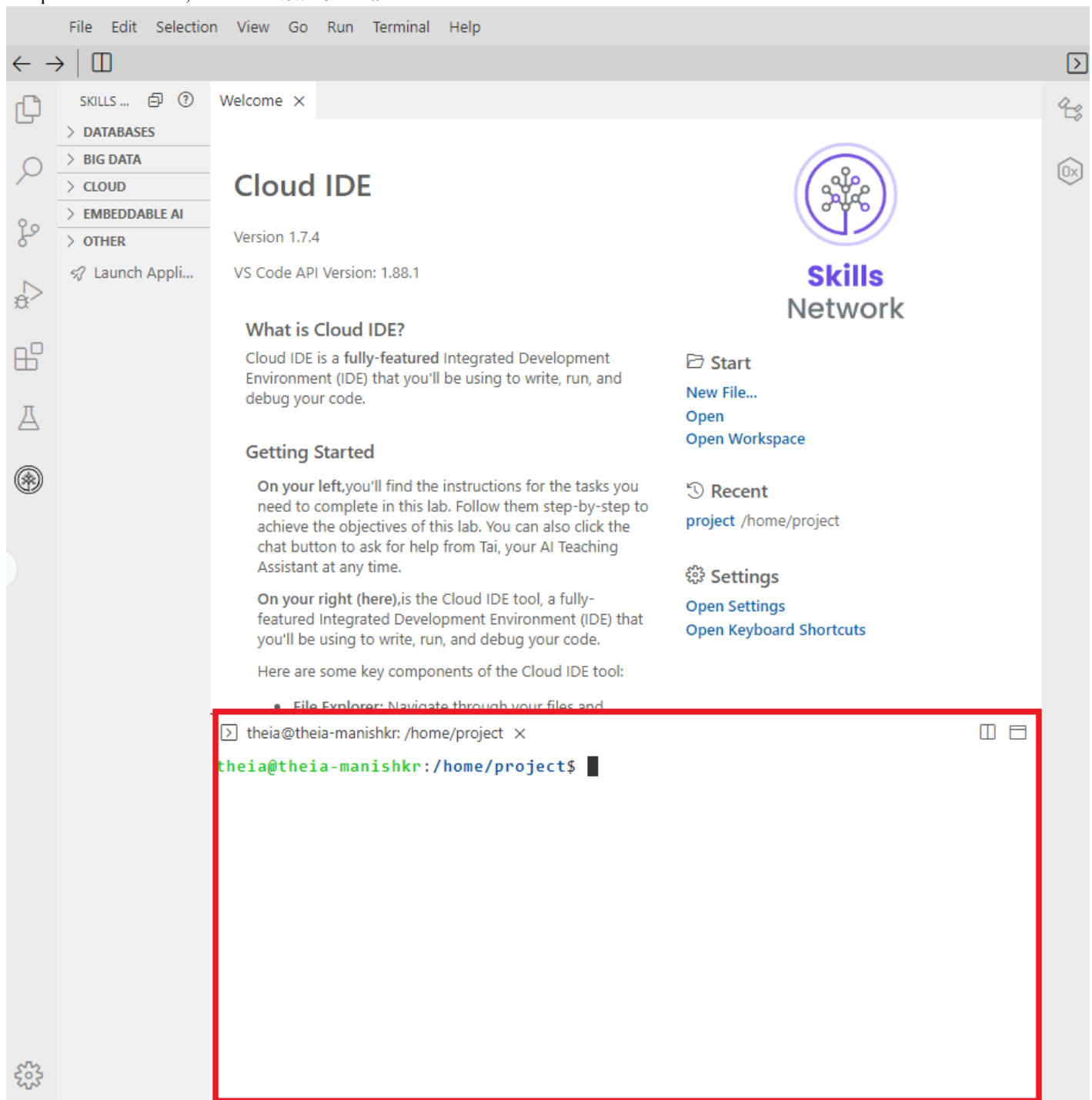
project/home/project

Settings

Open Settings

Open Keyboard Shortcuts

2. To open the new terinal, select the **New Terminal**



3. Copy and paste the commands into the terminal for the remaining steps.

Lab Setup

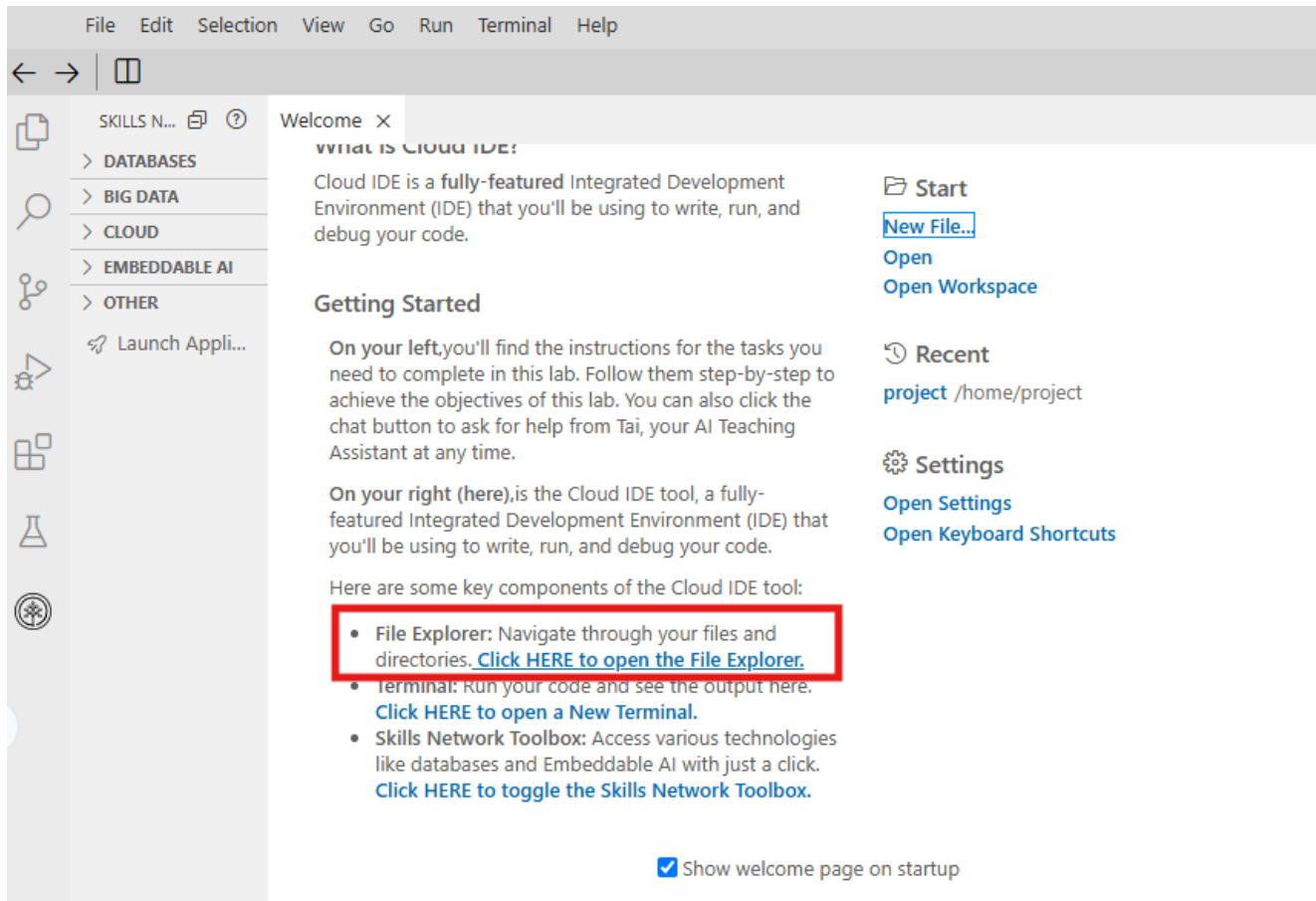
1. Install Steghide

Ensure steghide is installed on your system. If not, install it using:

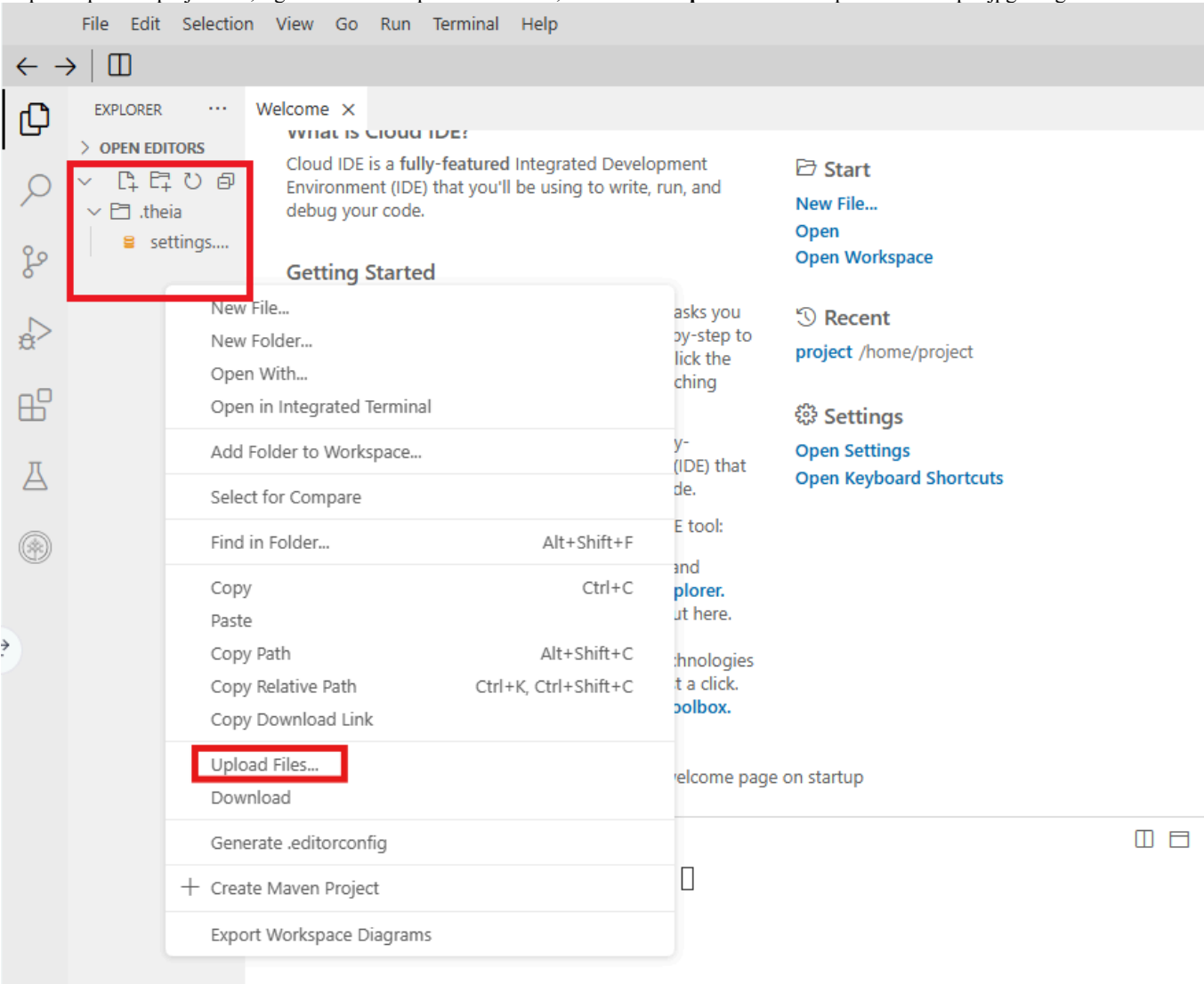
```
sudo apt update
sudo apt install steghide
```

2. Prepare sample files
Rename any image file on your local system to `sample.jpg` and upload it.
3. Follow the below steps to upload the file

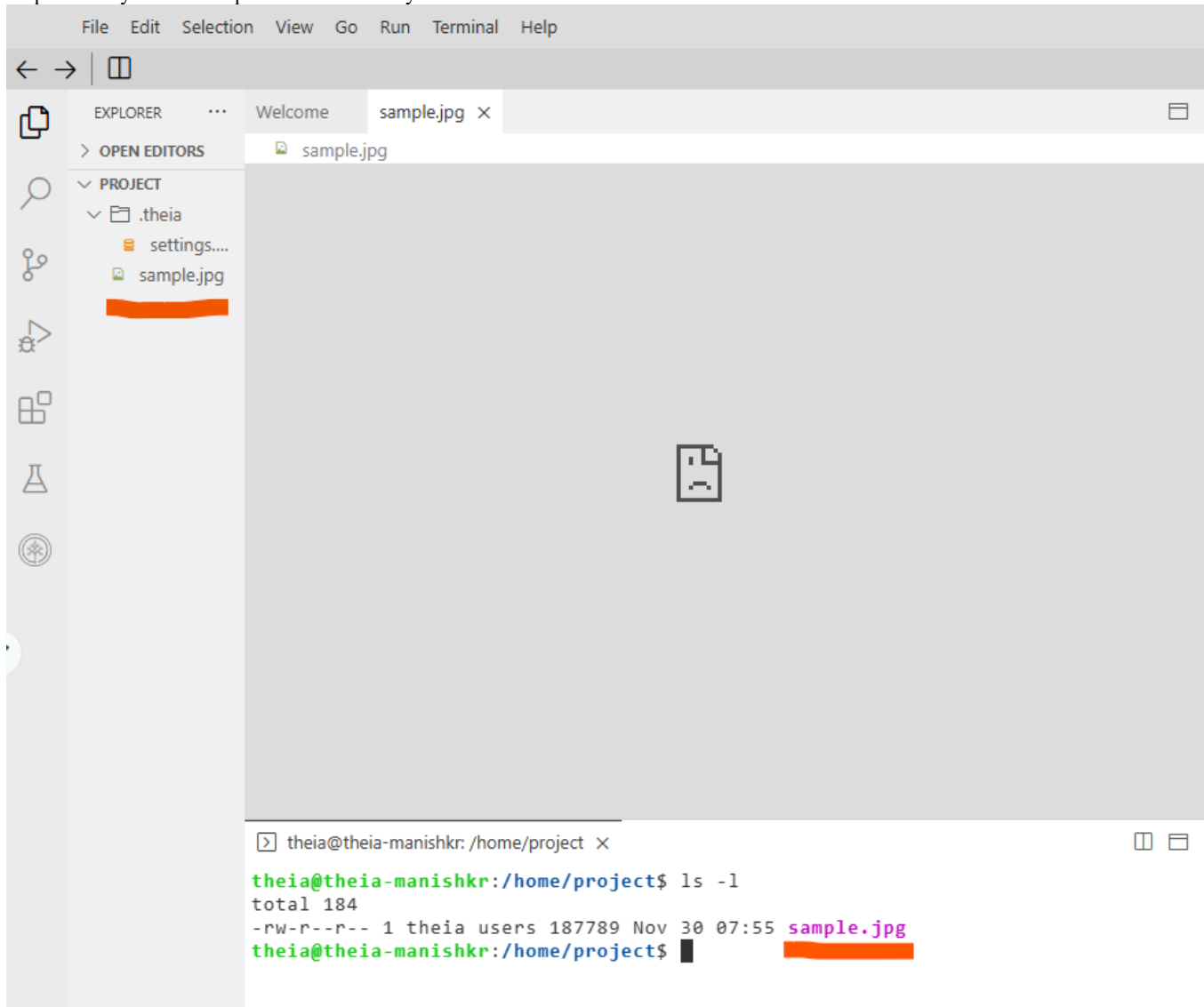
Step A: Select **File Explorer**



Step B: Open the project tab, right-click on the panel beneath it, and select /‘Upload File/’ to upload the sample.jpg image file.



Step C: Verify the file is uploaded successfully



Step D:- Create a text file secret.txt

```
echo "This is a secret file" > secret.txt
```

Step 1: Check File Compatibility

Verify whether a file is compatible for steganography with steghide using the below command.

```
steghide info sample.jpg
```

Command breakdown:

- **steghide info**: Displays information about the carrier file.
- **sample.jpg**: The carrier file being checked.

Expected output:

Information about the file's suitability for embedding data, such as available capacity.

Step 2: Hide a Message

Embed a secret message into the carrier file using the below command.

```
steghide embed -cf sample.jpg -ef secret.txt
```

Command breakdown:

- **embed**: Specifies embedding mode.
- **-cf sample.jpg**: Indicates the carrier file.
- **-ef secret.txt**: Specifies the file containing the secret message.

Prompt:

You will be prompted to enter a passphrase. Choose a strong password.

Step 3: Extract the Hidden Message

Retrieve the hidden message from the carrier file using the below command.

```
steghide extract -sf sample.jpg
```

Command breakdown:

- **extract**: Extracts the hidden file.
- **-sf sample.jpg**: Indicates the steganographic file to extract from.

Prompt:

You will be asked to provide the passphrase used during embedding.

Expected output:

The extracted file (for example, `secret.txt`) will appear in the current directory.

Step 4: Experiment with the Audio Files

Step A: Use an audio file as the carrier

Use the below command to replace the image file with an audio file (`sample.wav`) and repeat Steps 2 and 3. For example:

```
steghide embed -cf sample.wav -ef secret.txt
steghide extract -sf sample.wav
```

Step B: Play the audio file

Use the below command to play the audio file to ensure it remains functional.

```
aplay sample.wav
```

Exercises

Exercise 1. Hide Larger Files:

Try embedding a larger file (for example, `.pdf` or `.zip`) within an image or audio file.

Embed the larger file (for example, `secret.zip`) into an image (`image.jpg`) using the below command.

```
steghide embed -cf image.jpg -ef secret.zip -p mypassword
```

Hinted output:

```
Enter passphrase: *****
Re-Enter passphrase: *****
embedding "secret.zip" in "image.jpg"... done
```

Note: File sizes increase slightly before and after embedding.

Exercise 2. Compare File Sizes:

Observe how embedding data affects the file size of the carrier file.

Verify file size changes

```
ls -lh image.jpg
```

Hinted output:

```
-rw-r--r-- 1 user user 1.2M Nov 27 12:34 image.jpg
```

Exercise 3. Experiment with other tools

Use alternative Linux tools for steganography, such as outguess or stegsnow.

1. Outguess:

```
sudo apt install outguess
```

2. Stegsnow (for text-based files):

```
sudo apt install stegsnow
```

Conclusion

In this lab, you explored the basics of steganography using steghide on Linux. You successfully embedded and retrieved hidden data in carrier files and analyzed steganographic content. These techniques are fundamental for understanding how data can be correctly transmitted.

Author(s)

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